

µP Interfacing and Programming (EE3002)

Date: 22nd December, 2025

Course Instructor(s)

Dr. Muhammad Akmal

Ms. Abeer Bashir

Final Exam

Total Time (Hrs): 3
Total Marks: 75
Total Questions: 5

Roll No _____

Section _____

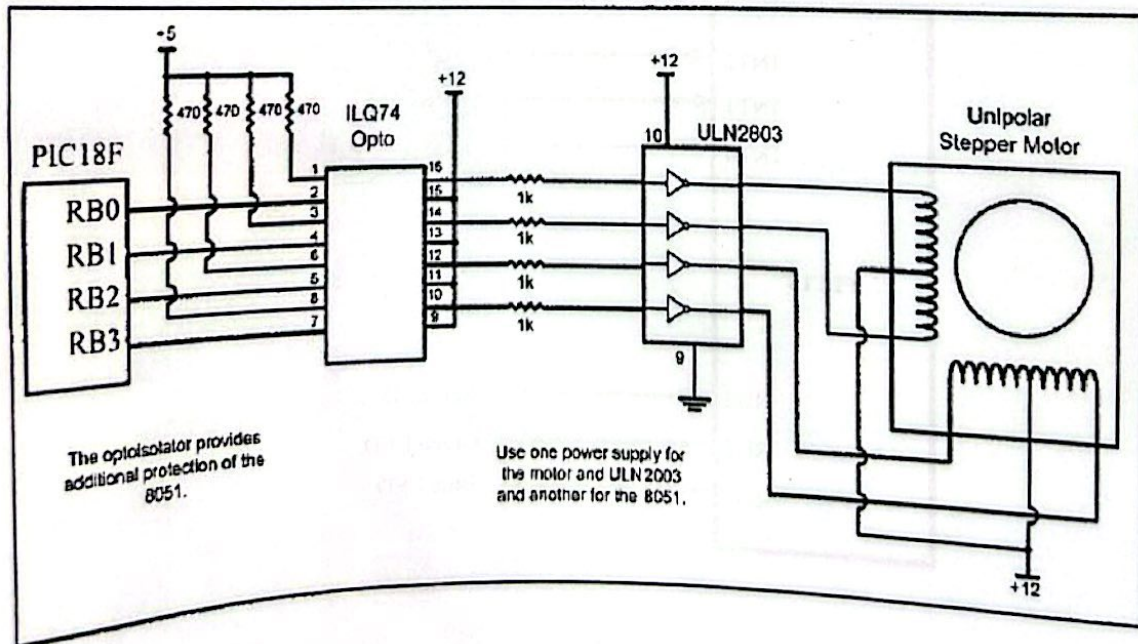
Student Signature _____

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- Instructions/Notes:
1. Attach the question paper with answer sheet.
 2. Some registers are given at the end of question paper.
 3. Make assumptions if you think some data is missing.
 4. Solve on the answer sheets. Answers written on the question paper will not be marked.

CLO 3: Construct a program to interface a microcontroller with external and internal components
(PLO 3, C6)

Q1: Unipolar stepper motor is connected to PortB of PIC18F as shown in the figure below. **Construct** an assembly language program (ALP) to rotate the motor clockwise. Use Timer0 to generate maximum possible DELAY in between executing each rotation sequence. Assume XTAL = 10 MHz, no prescaler, 16-bit mode. [Marks 15]



CLO 3: Construct a program to interface a microcontroller with external and internal components (PLO 3, C6)

Q2: Construct a C program to configure the serial communication module of a PIC18F to continuously transmit your name serially at a baud rate of 57600, using 8-bit data format and 1 stop bit. Assume XTAL= 10MHz. [Marks 10]

CLO 4: Design a system using micro-processor and micro-controller (PLO 3, C6)

Q3: Design a PIC18F-based stepper motor control system by writing an ALP, where: [Marks 20]

1. Motor speed is controlled using a timer interrupt
2. Motor direction is changed using an external interrupt
3. Current speed, direction, and fault status are displayed on an LCD

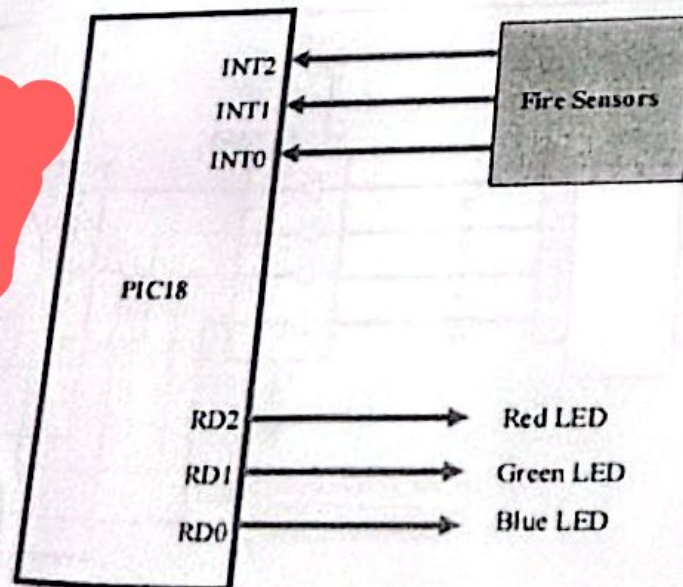
Justify your choice of timer, interrupt priorities, and LCD update strategy to avoid missed steps.

CLO 4: Design a system using micro-processor and micro-controller (PLO 3, C6)

Q4: Develop an ALP for the system shown in the figure, using a PIC18F microcontroller, to configure multiple external interrupts generated by three fire sensors connected to INT0, INT1, and INT2. Your program should demonstrate the given logic: [Marks 20]

- a) Blinks Red LED when interrupt occurs on INT0.
- b) Blinks Green LED when interrupt occurs on INT1.
- c) Blinks Blue LED when interrupt occurs on INT2.

NOTE: Ensure that when INT0 interrupt is triggered, only Red LED blinks, while the Blue and Green LEDs should be OFF. Same for the other cases.



CLO 3: Construct a program to interface a microcontroller with external and internal components (PLO 3, C6)

Q2: Construct a C program to configure the serial communication module of a PIC18F to continuously transmit your name serially at a baud rate of 57600, using 8-bit data format and 1 stop bit. Assume XTAL= 10MHz. [Marks 10]

CLO 4: Design a system using micro-processor and micro-controller (PLO 3, C6)

Q3: Design a PIC18F-based stepper motor control system by writing an ALP, where: [Marks 20]

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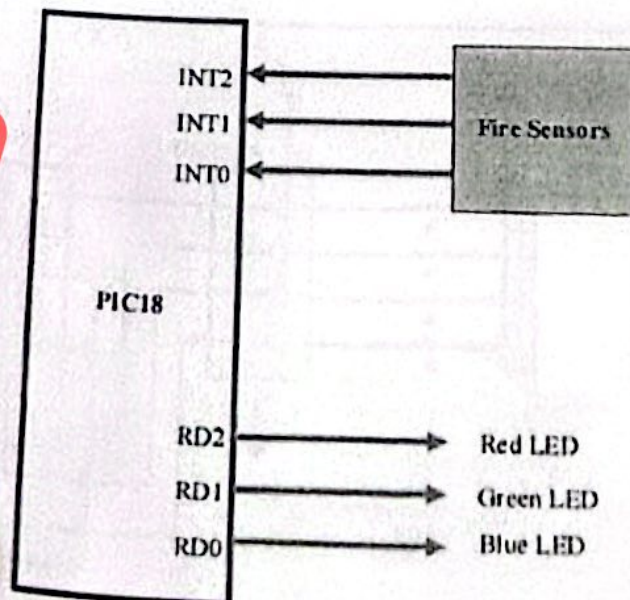
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CLO 4: Design a system using micro-processor and micro-controller (PLO 3, C6)

Q4: Develop an ALP for the system shown in the figure, using a PIC18F microcontroller, to configure multiple external interrupts generated by three fire sensors connected to INT0, INT1, and INT2. Your program should demonstrate the given logic: [Marks 20]

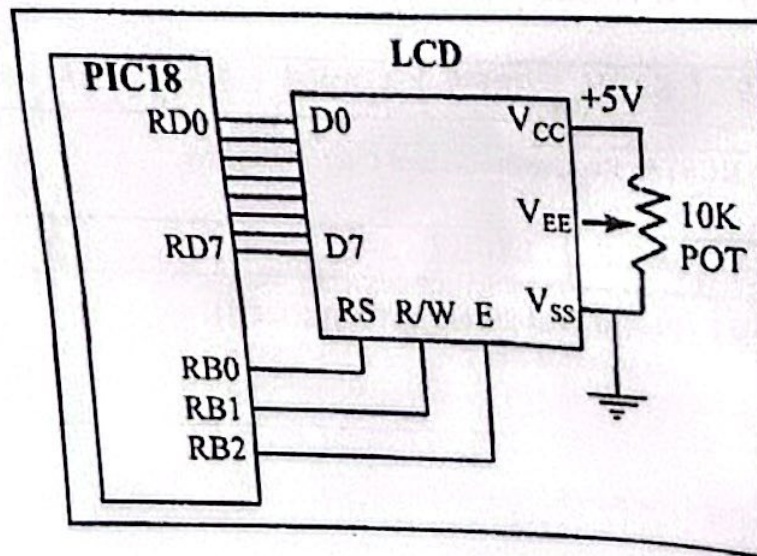
- a) Blinks Red LED when interrupt occurs on INT0.
- b) Blinks Green LED when interrupt occurs on INT1.
- c) Blinks Blue LED when interrupt occurs on INT2.

NOTE: Ensure that when INT0 interrupt is triggered, only Red LED blinks, while the Blue and Green LEDs should be OFF. Same for the other cases.



CLO 3: Construct a program to interface a microcontroller with external and internal components
(PLO 3, C6)

Q5: Construct an ALP for the PIC18F to display your Roll number on LCD. Use the configuration shown in the figure.
[Marks 10]



TXREG register RCREG register

CSRC	TX9	TXEN	SYNC	0	BRGH	TRMT	TX9D
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TXSTA: Transmit Status and Control Register

SPEN	RX9	SREN	CREN	ADDE	FERR	OERR	RX9D
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RCSTA: Receive Status and Control Register

---	--	RCIF	TXIF	--	---	--	
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PIR1 (Peripheral Interrupt Register 1)